

Adrien Goldszal

curriculum vitae

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Education

- 2024–present **Diplôme d'ingénieur**, *Ecole Polytechnique*, Palaiseau, France
Third year specialisation in artificial intelligence and data science at one of France's leading scientific institutions, ranked best engineering *Grande Ecole* in the country.
Relevant coursework includes : *Machine Learning and Autonomous Agents*, *Image Analysis and Computer Vision*, *Reliable Collaborative Learning*
- 2022–2024 **Gradué d'ingénierie - Bachelor level**, *Ecole Polytechnique*
Major in Advanced Science and Engineering, focusing on computer science & applied mathematics. Additional courses in mechanics and physics.
Relevant coursework includes : *Optimisation and Control*, *Algorithms for Data Analysis in C++*
GPA 3.76/4.0
- 2020–2022 **French Preparatory Classes**, *Lycée Privé Sainte Geneviève*, Versailles, France
Intensive two-year preparation in mathematics, physics and chemistry for the entrance exams of France's best *Grandes Ecoles*.
Undergraduate coursework includes *Linear Algebra*, *Real Analysis*, *Solid and Fluid Mechanics*
GPA 3.89/4.0
- 2017–2020 **Baccalauréat - Abitur**, *Lycée International*, Saint-Germain-En-Laye, France
Double French and German Baccalauréat and Abitur.
19.27/20 (1.0) Highest Honours

Work Experience

- April-Aug. **Incoming Research Intern**, *MILA - Quebec AI Institute*, Montréal, Canada
2025 5-month research project under Professor Pierre-Luc Bacon. I will be working on scalable and uncertainty-aware neural stochastic differential equations for optimal control of real world systems, leveraging trajectory flow matching and direct epistemic uncertainty prediction.
- June-Aug. **AI Research Intern**, *Siemens Technology*, Munich, Germany
2024 Three-month internship in the Digital Manufacturing Technologies research group under the supervision of Dr. Peter Robl
Deep reinforcement learning (RL) for automated electrical cable routing in continuous 3D space.
 - Implemented a search heuristic and leveraged demonstrations to improve the RL frameworks' performance
 - Used Explainable AI techniques like Shapley Values to optimize agent observations
 - Developed an API for Siemens clients facilitating the use of the lab's cable routing tools
- 2023–2024 **English Oral Examiner**, *Lycée Privé Sainte Geneviève*, Versailles, France
Preparing multiple classes of students for the english oral tests of the entrance exams for the top French *Grandes Ecoles* through one-on-one tutoring and training.

- 2022–2023 **Officer Cadet & Assistant Project Manager**, *Southeastern France Regional Fire Department*, Bourg-en-Bresse, France
Ecole Polytechnique first year military internship (Six months).
- Worked as a first responder, performing first aid & vehicle extractions.
 - Developed new and optimized tools to monitor and regulate the service's energy consumption
 - Coordinated a plan to reduce the service's carbon emissions by 30% (60 fire stations, 4000+ people)

Selected Research & Project Work

Research

- 2024–present **Multi-Agent RL for Drone Swarm Rescue Operations**, *Ensta Paris / Inria FLOWERS*
One year part-time research project under the supervision of Dr David Filliat, robotics director at Ensta Paris.
- Implementing and testing multi-agent reinforcement learning algorithms in a drone swarm rescue environment.
 - Leveraging state of the art agent communication techniques as well as curriculum learning to improve performance.

Course-related : Computer Science

- Nov.–Dec. 2024 **Cooperative Inverse Reinforcement Learning (CIRL)**, *Ecole Polytechnique*
Conducted an in-depth analysis and implementation of two research papers on CIRL in the specific case of apprenticeship learning (Dylan Hadfield-Menell et al. Neurips 2016), and in the general CIRL framework (Dhruv Malik et al. ICML 2018) for Ecole Polytechnique's *Collaborative & Reliable Learning* course.
- Nov.–Dec. 2024 **Automatic Puzzle Solver**, *Ecole Polytechnique*
Two-month project for Polytechnique's *Image Analysis and Computer Vision* course.
- Developed an automated real-time puzzle-solving system leveraging SIFT-based feature detection, edge detection, and morphological operations for robust piece identification
 - Enhanced accuracy in puzzle solving through manual calculation of homography transformations and validating placements using Zero-mean Normalized Cross-Correlation (ZNCC)
- Feb.–March 2024 **3D Animated Scene**, *Ecole Polytechnique*
Two-month project for Polytechnique's Computer Graphics Course.
- Created a 3D animated scene in an open environment using the OpenGL library.
 - Modeled and integrated real-time physics, resulting in a more realistic and immersive user experience and implemented user commands for interactive control.

Course-related : Engineering

- 2023–2024 **Atmospheric Balloon**, *Ecole Polytechnique Spatial Center*
One year group engineering research project in collaboration with the LATMOS (French National Meteorology Lab).
- Created new adapted tools and sensors sent on an atmospheric balloon for the study of multiple physical phenomena including conductivity and the electrical field of the earth
 - Developed the code for efficient data acquisition, transmission and analysis.
- 2021–2022 **Study of sand dune movement**, *Lycée Sainte-Geneviève*
One year group scientific research project for the entrance examinations of the French *Grandes Ecoles*.
- Designed and built an experimental water tank system with automated sand displacement mechanisms to simulate desert dune formation at 1:1000 scale.
 - Implemented a Python-based cellular automaton model incorporating fluid dynamics principles, successfully predicting dune migration patterns.

Activities & Volunteering

2023–Present **President & Software Team Leader, IP Paris Racing Team**

IP Paris Racing Team is an association created in 2023 to represent the Institut Polytechnique de Paris at Formula Student: designing an electric car to participate in international student competitions.

- Coordinating a team of 50 students, raising funds and finding sponsors for IP-Paris' first Formula Student vehicle
- Leading a team of 7 on the electronics of the car and the associated software. Working on the code for the communication between sensors in embedded C.

2024–Present **Core Member, X – AI (Polytechnique AI Association)**

X-AI helps promote and motivate students wanting to discover more about the AI sector.

- Organized multiple hackathons and conferences with leading AI research firms such as Google DeepMind & QuantumBlack
- Held introductory courses and tutoring sessions for first- and second-year students

Languages

French	Native	<i>French Baccalauréat (2020)</i>
German	C2 - Proficient	<i>German Abitur (2020)</i>
English	C2 - Proficient	<i>Cambridge Proficiency Exam : Grade A (2020)</i>
Spanish	B2 - Intermediate	

Technical Skills

ML/AI Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV, OpenAI Gym
Data Engineering	SQL, Spark, Pandas, NumPy
Programming	Python, C++, Java, Git
Hardware	PCB Design, Arduino, Sensors